

NEURAX HACKATHON 3.0

DOMAIN 2 · AI IN INDUSTRY AND AUTOMATION

Problem Statement: *Visual Inspection & Defect Root-Cause Assistant*

Description:

High-throughput manufacturing lines are difficult to optimize because product quality, process capacity and economics interact. Defect signatures can be subtle and easy to miss at production speed, while bottlenecks may emerge from cycle-time imbalance, excess work-in-process, downtime, changeovers, low utilization, scrap or rework. For this challenge, teams should design for a realistic multi-stage manufacturing environment with mixed product variants, changing inspection conditions, recurring defect families, batch-to-batch process drift, varying station capacity and operating conditions that affect both throughput and profitability.

The objective is not to build another image classifier, generic labeling tool or isolated KPI dashboard. The objective is to build a unified industrial decision-support system that can explain what is wrong with a product, where the defect sits, what process condition is likely responsible, where the production flow is constrained, how the constraint affects throughput and cost, and how quality or process changes are likely to affect profitability.

Core Problem Statement:

Build a software-only AI system that analyzes organizer-provided inspection, production and economic datasets to produce a continuously updated view of product quality, production-flow health and expected profitability. The system should classify units as acceptable or defective, localize defects where the data supports it, flag uncertain or novel defect types instead of forcing a guess, identify industrial bottlenecks, estimate their effect on throughput and losses, predict profitability or margin, and generate evidence-based process recommendations.

All recommendations, bottleneck interventions and profitability estimates must remain simulated or advisory. No live camera feed, PLC connection, robotic sorting integration, machine control or production-line hardware access is required or permitted for judging.

Complete Evaluation — 100 Marks

- CHECK POINT 1 – 15 MARKS**

Criterion	Marks	What Judges Should Look For
README	15	Problem Understanding (5), Architecture (5), approach (5).

- CHECK POINT 2 – 25 MARKS**

Criterion	Marks	What Judges Should Look For
Partial Execution	25	Features, Problem Relatability.

- **CHECK POINT 3 – 60 MARKS**

Criterion	Marks	What Judges Should Look For
Defect detection & classification accuracy	15	Correctly identify defective units and defect categories while controlling false rejects.
Defect localization quality	10	Precision of boxes, masks or heatmaps on unseen samples.
Robustness to unseen conditions	10	Performance under lighting/orientation/batch changes and novel variants.
False-reject / false-accept handling	5	Balance between missed defects and unnecessary rejections.
Root-cause correlation quality	5	Validity of links between defect patterns and process/batch data.
Explainability & confidence handling	5	Evidence, calibrated confidence, uncertainty and limitations.
Technical implementation	5	Architecture, reproducibility and runtime reliability.
UI/UX & visualization	5	Operational clarity of dashboard, trends and evidence.